9 **(a)** Carbon-14 is the only radioactive isotope of carbon that occurs naturally.

(i) State **one** similarity and **one** difference in the composition of the nuclei of atoms of different isotopes of the same element.

similarity .....

difference .....

[1]

(ii) When a tree dies, no new carbon-14 is absorbed by the tree.

A tree contains 16 μg of carbon-14 when it dies.

Carbon-14 has a half-life of 5700 years.

On Figure 9.1, sketch a graph to show how the mass of carbon-14 in the tree decreases during the first 15 000 years after the tree dies.

Show your working and label the axes with suitable values for **one** point on the graph.

Figure 9.1

[3]

(b) Complete the sentence about radiation.

Ionising nuclear radiation that is always present in the environment is known

as ..... radiation.

[1]

(c) State **one** source that makes a significant contribution to the ionising nuclear radiation that is always present in the environment.

..... [1]

[Total: 6]

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